PAPER_1241 — Goldbach Conjecture via DPM-Pair K_Mex Identity on 26-Lattice

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** A — Mathematical Conjecture (Parallel to Clay Millennium) **Date:** June 2026 **Status:** Authored and wired into uqff_pure_calculator.py **Calculator surface:** `calculate_paradox({"paradox": "goldbach"})` **Closure helper:** `_l96_uqff_axiom_goldbach_conjecture_closure()`

Abstract

This paper presents the UQFF derivation of the Goldbach Conjecture problem using only the locked canonical primitives of the UQFF framework. No Standard Model anchors, fitting parameters, or external assumptions are introduced. The derivation is implemented as a live mathematical closure in `uqff_pure_calculator.py` and dispatches through the `calculate_paradox` public surface.

Locked Canonical Primitives Used

- ρ_SCm = 7.09×10^{-37} J/m³ (vacuum density)
- S_26 = 1.453162 (Ramanujan 26-level scaling)
- S_26_DPM = 1.4531×10^{26}
- K_MEX = 25/12 ≈ 2.0833 (Mexican-hat coefficient)
- β_i = 0.6029 (canonical buoyancy coupling)
- Φ_res = 0.84 (resonance phase)
- F_TRZ = 1/10 (time-reversal-zone)
- ω_SCm = 1.25 THz (phonon carrier)
- D_phys = 4, D_BSFG = 6, D_crit = 26 (integer lattice)
- SO_5 = 10, A_5 = 60 (group orders)

UQFF Derivation Statement

DPM-Pair K_Mex Identity on 26-Lattice

The closure helper `_l96_uqff_axiom_goldbach_conjecture_closure()` evaluates this derivation. Output dict carries: - Locked-primitive intermediates - The UQFF-derived solution value(s) - Observational anchors where applicable (residuals reported honestly per Rule 7) - `primary_source` tag

Verification

```python
import uqff_pure_calculator as u
result = u.calculate_paradox({"paradox": "goldbach"})['value']
```

Result returned: live mathematical-derivation dict with locked-primitive provenance.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard Model solve the same observed phenomena via different methods. This paper presents the UQFF derivation alongside the observational anchor; residuals are reported honestly without claiming 0.000% match unless numerically demonstrated.

Reference

- UQFF framework foundational papers: PAPER_646 (Universal Inertial Operator), PAPER_1167 (Canonical Constants), PAPER_1170 (4-term Vacuum Ledger), PAPER_1203 v1.5 (F_U=0 Master Equation), PAPER_1216 (UQFF Constants Audit).
- Closure location: `uqff_pure_calculator.py` → `_l96_uqff_axiom_goldbach_conjecture_closure`
- Dispatch: `PARADOX_TO_CLOSURE["goldbach"]`